

Data-driven auction design for blockchain-based digital asset trading: A mixed method

Su Xiu Xu* Yifang Ding[†] Meng Cheng[‡] Sini Guo[§] Xiang T.R. Kong[¶]
George Q. Huang^{||}

Abstract

This paper is motivated by a real project with a leading culture assets and equity swap organization in South China. Our preliminary survey assesses the potential of digital collectibles swap, with professionals aiming to boost liquidity and consumers driven by social attributes and meta-item diversity. In our setting, each agent owns a digital asset and wants another meta-item. However, the traditional *Vickrey-Clarke-Groves* (VCG) auction runs at a deficit. We thus introduce a novel mechanism that combines the VCG auction with *limited supply* and *platform escrow* concepts, called *LSE-VCG auction*. To improve the surplus of platform (auctioneer), we use *limited supply* to constrain the number of winners and *platform escrow* to increase market demand. The LSE-VCG auction satisfies both truthful telling and participation rationality. If multilateral matching achieves maximal social welfare, then the substitute condition does not hold (*impossibility theorem*). We prove that the platform's surplus can be improved by limited supply in some conditions. Our experimental results show that the VCG auction solely with limited supply could reach greater social welfare, agents' profits and ratio of swap relative to the sequential Vickrey auctions. Moreover, a mix of limited supply and platform escrow schemes can further improve the platform's profit and successful trading ratio. Truthful telling is almost an optimal strategy for the platform, thus promoting a transparent and beneficial auction environment. Besides, the impacts of auction timing, market size, value distribution and size of XOR bids are investigated. Finally, we apply an effective machine learning method to predict the limited supply number with partial information before the auction.

Keywords: Metaverse swap; Auction design; Platform escrow; Limited supply; Incentive compatibility; Data-driven methods.

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1 Introduction

The metaverse is a virtual space full of items, interconnected with the physical world via network infrastructure (Huang et al., 2023; Orazi and Nyilasy, 2025). According to a recent report from Citi, the metaverse market size rises up to \$13 trillion (about half of US's GDP) by 2030¹. Leading tech firms like Meta, Google, Microsoft, and Nvidia have invested heavily in advancing metaverse technology. The key manufacturers like BMW and Siemens are leveraging metaverse concepts to innovate in supply chain (Dolgui and Ivanov, 2023). Fashion brands including Puma, Nike, Gucci, and Adidas are exploring metaverse marketing and e-commerce to enhance psychical and digital product sales (Holzwarth et al., 2022; Bellezza, 2022). Furthermore, the metaverse enjoys vital potential for applications in various fields such as games, education and healthcare (Yoo et al., 2023). As said by Mark Zuckerberg, the metaverse will enable the public and firms to shop, entertain, and trade virtual assets through digital platforms², for example, the Roblox, Decentraland, Cryptovoxels, and OpenSea. Therefore, with the booming of the digital market platforms, more and more users engage in and list their digital assets, set prices, and conduct transactions.

Metaverse commerce can be of two kinds: verification of ownership rights and pricing. The verification of ownership rights includes, for example, shifting full and fractional ownership through the uniqueness technologies like blockchains (Colicev, 2023). Effective pricing mechanisms are essential for ensuring the long-term stability of the virtual asset trading market within the metaverse. The prevalent pricing methods currently include fixed pricing and floating pricing. Generally, fixed pricing is adopted in mature and stable markets, while floating pricing is a better strategy when the supply and demand in the market are relatively uncertain or subject to significant fluctuations and randomness. Apparently, the digital asset trading market is still in its developmental phase, with relatively random supply and demand. In this paper, we view the issue of digital asset trading as a floating pricing problem.

This study is motivated by a real project with a leading culture assets and equity swap organization (called organization X) in South China. The organization launched a digital asset trading platform in August 2022, which is designed to unlock the full potential of digital assets and has reached a milestone of cumulative transactions of 20 billion RMB by February 2024. As illustrated in Fig. 1, the digital assets (such as cultural IP, digital copyrights and collectibles) are recorded on different consortium blockchains, enabling secure ownership, interoperable transfers, and smart contract programmability. The platform facilitates digital asset trading via English auctions, fixed pricing, and negotiated deals, generating revenue through commission charges. Both institutional entities and private individuals can engage as buyers/sellers in digital asset transactions, which naturally forms four business models, i.e., B2B (business-to-business), B2C (business-to-customer), C2B (customer-to-business) and C2C (customer-to-customer). However, the C2C digital collectibles market remains in its infancy, characterized by limited trading liquidity. Currently, the challenges faced by organization X are twofold. Firstly, to enhance the market liquidity of digital collectibles, the establishment of a new trading mechanism is imperative. Moreover, Organization X relies on traditional English auctions, which yield only

¹<https://www.precedenceresearch.com/metaverse-market>

²<https://www.cbsnews.com/news/metaverse-facebook-virtual-augmented-reality-hubs/>

modest commission-based revenues. Thus, organization X wants to try other auction mechanisms, which could bring higher profits when guaranteeing market liquidity. In our collaborative project, organization X demonstrates openness to innovative mechanism design so as to boost digital asset liquidity and platform revenue.

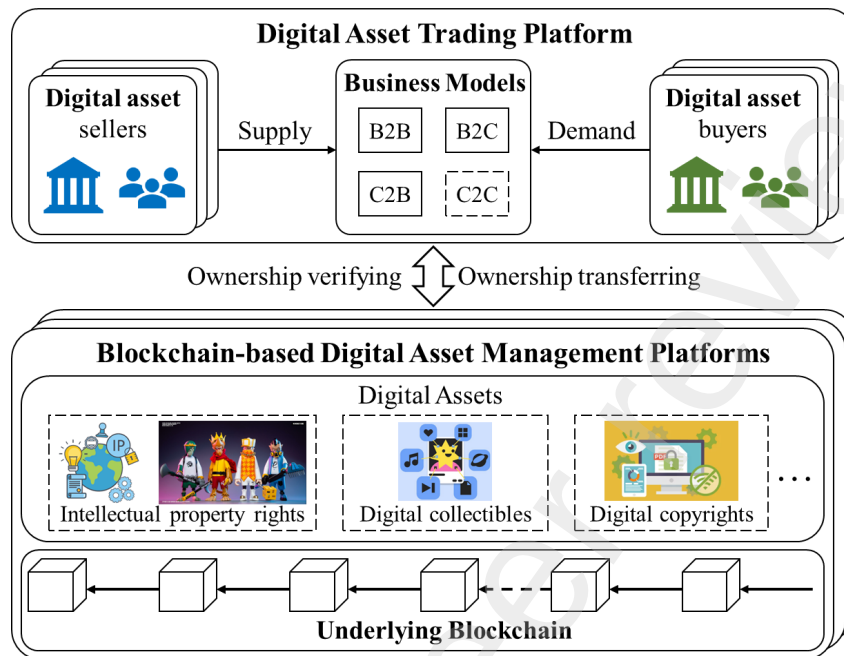


Fig. 1. Blockchain-based digital asset trading ecosystem

To date, there have been two types of floating pricing mechanisms in the general metaverse environment. One is to release the price of digital assets based on real-time index (e.g., bitcoins), the other is to use incentive schemes like auctions. A recent review by Huang et al. (2023) summarizes some classic auction mechanisms for the allocation problem within the metaverse, such as the English, Dutch, Vickrey, and double auctions. However, such traditional auction mechanisms cannot be used in our setting, in which each agent is simultaneously a buyer and a seller of digital assets. As in our questionnaire statistics (see Section 3.1), 96.20% of institutional professionals are willing to participate in digital collectible swaps, primarily to acquire higher-value assets and enhance market liquidity. Also, 79.57% of consumer respondents are motivated more by social connectivity and meta-item diversification. Indeed, our proposed auctions leverage the underlying idea of well-known Vickrey-Clarke-Groves (VCG) mechanism (Vickrey, 1961; Clarke, 1971; Groves, 1973). In the VCG auction, each agent's profit is his marginal contributions to the allocation system, leading to incentive compatibility. The family of VCG auctions has been applied to urban freight management (Yang et al., 2019), edge computing transaction in the metaverse (Luong et al., 2024) and metaverse-enabled logistics procurement (Xu et al., 2025). Unfortunately, the direct use of VCG auction will cause budget deficit in the presence of multilateral swap/matching.

In the same time, there are two main learning methods in auction mechanisms: (i) learning in repeated auctions, in which a bidder can learn their own or others' valuations over time (Kim and Koh, 2020; Daskalakis and Syrgkanis, 2022; Bobkova, 2024; Han et al., 2025); (ii) learning in one-shot auctions, where the equilibrium solutions (Bichler et al., 2023) and model parameters

(Kim and Choi, 2024) can be predicted by using machine learning or neural network. However, such prediction methods cannot be directly applied to our multilateral swap setting. To mitigate budget deficit, an appropriate parameter could be incorporated in metaverse swap auctions. But this optimal parameter can only be determined after the auction, once all information is available. The challenge lies in predicting this parameter before the auction with partial information.

In this paper, we aim to develop effective auction mechanisms for metaverse multilateral trading. Specifically, we will answer the following questions: (i) How to design incentive-compatible and budget-balanced auctions with high efficiency? (ii) What are the impacts on the multilateral market if agents can sell meta-items to a third-party platform? (iii) What are the impacts of negative valuations and other realistic factors (e.g., a selfish platform, auction batches, market size)? (iv) How to compare the performances of different auction mechanisms? (v) How to prevent budget deficits for the third-party auction platform using machine learning?

We consider a digital asset swap setting with multiple agents and a third-party platform. The platform/auctioneer (e.g., Foundation NFT company, organization X) runs an auction and the agents are bidders. Each agent possesses a digital asset and is interested in acquiring the ownership of an additional digital asset. We prove that if the multilateral matching achieves maximal social welfare, then the substitute condition does not hold (*impossibility theorem*). To improve the platform’s revenue, we use the concept of *limited supply*, which specifies an upper bound on the quantity of winners. Indeed, the concept of *limited supply* serves as a form of *competition enhancement* (Chu and Shen, 2006, 2007, 2008; Chu, 2009; Roughgarden et al., 2020), as scarcity can stimulate higher revenue outcome. Note that, a seminal paper by Chu and Shen (2006) develops an agent competition double-auction mechanism with transaction costs in the single-unit trading environment. Regarding the importance of limited supply, we prove that the platform’s revenue is marginally greater if the maximum number of winners is equal to $|N| - 1$, where $|N|$ is the number of bidders. Furthermore, each agent can sell the usage right of his digital asset to the platform by receiving a payment, which is called *platform escrow* scheme in this paper.

By combining the concepts of limited supply and platform escrow, we propose a novel VCG-like auction mechanism. Such auction has good properties like incentive compatibility and individual rationality. Based on experimental tests, we gain some interesting results and insights. First, compared to the VCG auction with limited supply (LS-VCG auction), our mechanism with limited supply and platform escrow (LSE-VCG auction) yields greater social welfare and possibility of successful trading, as well as platform’s surplus. Moreover, the two auctions could reach greater social welfare, agents’ profits and ratio of swap relative to the sequential Vickrey auctions. According to the Revenue Equivalence Principle (Myerson, 1981; Riley and Samuelson, 1981; McAfee and McMillan, 1987; Maskin et al., 1989), the Vickrey auction is equivalent to that of any standard auction with risk-neutral and symmetric bidders, such as English auction, Dutch auction and the first-price sealed auction. Second, although the platform may benefit from a selfish behavior in the theoretic level, our simulations find that truthful telling is almost an optimal strategy. The result is inconsistent with the combinatorial VCG auctions with one-sided matching (Xu et al., 2018). The reason is that there are some

risks associated with overstating or understating in the multilateral swap environment. Third, frequent auctions can achieve budget balance earlier than running auction seldom as the limited supply number reduces. The higher the number of auction batches is, the larger the limited supply number required to reach positive surplus for the auctioneer. Fourth, we demonstrate the effectiveness of platform escrow in extensive experimental environments. Finally, we propose using machine learning methods to predict the limited supply number with partial information before the auction. The result indicates that utilizing the predicted limited supply number can achieve high platform revenues and retain a large number of participants in a long run.

Our work develops novel VCG-like auctions for metaverse multilateral swap by introducing and enriching the concepts of limited supply and platform escrow. We also extend the auction theory upon promoting expected revenue with augmented demand and limited supply (Chu and Shen, 2006, 2007, 2008; Chu, 2009; Roughgarden et al., 2020; Chu and Wu, 2023). The managerial implications of our findings are as follows. First, the design of metaverse swap mechanisms can be based on the underlying idea of VCG auctions, as well as the schemes and machine learning methods that can achieve budget balance. Second, the platform needs to tell the truth for enhancing revenue and attracting more agents in the metaverse market. Third, conducting auctions more frequently enables earlier achievement of budget balance compared to infrequent auctions, as the limited supply parameter diminishes. Increased auction frequency necessitates a higher limited supply value to attain budget surplus. Finally, when designing metaverse swap mechanisms, the platform (auctioneer) should have a full understanding of the benefits and risks related to metaverse float pricing problems.

This paper is organized as follows: Section 2 reviews relevant literature, while Section 3 provides a preliminary survey and a LSE-VCG auction example. The formal LSE-VCG model and analysis are developed in Section 4. Section 5 discusses the simulation outcomes and conducts a comparative analysis of the LSE-VCG and the LS-VCG auctions. Section 6 presents the conclusions and the future research.

2 Literature review

There are two related streams of research upon metaverse operations management and economic system, as well as auction mechanism design. The first stream of literature concerns issues in metaverse operations management and economic system. Yoo et al. (2023) propose a novel conceptualization of the metaverse and outline research directions by exploring how metaverse dimensions amplify digital customer touchpoints, such as economic exchange and complex social relationships. Likewise, Dolgui and Ivanov (2023) introduce a framework for metaverse Supply Chain Operations Management (SCOM) and emphasize the potential of revolutionizing decision-making in areas such as inventory management, production planning, and digital pricing. Metaverse technologies may be capable of predicting consumers' purchase preferences with desirable precision levels (Hadi et al., 2024). Based on experimental studies, Orazi and Nyilasy (2025) show that metaverse-first retail strategies lower willingness to pay due to reduced investment value, through in-store atmospherics can mitigate the valuation penalty. A recent empirical study by Harz et al. (2022) indicate that virtual reality enhances the accuracy

of forecasts regarding the adoption of new products. In this paper, we also consider the business relationship between metaverse consumers and platforms. For example, the platforms can recycle the metaverse assets such as digital painting, virtual pet, virtual game access rights and digital property by effective pricing schemes.

Our work is also related to the metaverse economic system. A central concern is the pricing dynamics of metaverse assets. Zhang (2023) examines cryptocurrency and non-fungible tokens (NFTs) pricing and emphasizes that the value of Bitcoin stems from collective belief, which leads to volatility, while NFT prices depend on individual willingness to pay. Colicev (2023) argues that NFTs enable brands to build highly engaging communities, integrate online and offline ownership experiences, and strengthen emotional connections between brands and consumers. By analyzing large-scale field data, Yang (2024) further reveals that the consumers are inclined to pay more for the unused version of a digital good. In contrast, our work mainly explores the transactions among the holders of metaverse assets and their interactions with platforms through *consignment sales*.

The second stream of literature focuses on incentive mechanisms. Auctions are regarded as efficient incentive mechanisms for the valuation and swap of virtual goods and services. A recent review by Huang et al. (2023) summarizes some classic auction mechanisms for the allocation problem within the metaverse, such as the English, Dutch, Vickrey, and double auctions. In this paper, our proposed auctions are based on the underlying idea of Vickrey-Clarke-Groves (VCG) auction with multilateral swap. The VCG auction, noted by Varian and Harris (2014), has the appealing property that bidding one's true value is a dominant strategy for all players, with expected revenue equivalent to that of the Generalized Second Price (GSP) auction. In terms of collusive bidding, Decarolis et al. (2020) demonstrate that the VCG mechanism outperforms the GSP auction in terms of revenue and efficiency. In the context of multi-unit demands, Yan (2021) develops a two-stage ascending clock auction that dynamically implements the VCG outcome. The family of VCG auctions has been applied to the storage allocation in video surveillance camera systems, edge computing transaction in the metaverse (Luong et al., 2024) and e-commerce logistics (Xu and Huang, 2017; Zhang et al., 2019). Despite the merits of VCG-like auctions, there are some challenges in practical implementations. For example, sub-optimal outcomes can yield non-truthful VCG-based mechanisms (Nisan and Ronen, 2007). Regarding the complex allocation problems, Niazadeh et al. (2022) suggest that the VCG auction might yield lower revenue and propose a core pricing algorithm which generates GSP auction matched revenue. For a comprehensive review of the evolution of auction theory and its applications in practical market design, readers can refer to Milgrom (2021).

In our meta-item swap setting, each agent is *simultaneously* a buyer and a seller. However, the direct application of VCG auction in our scenario will probably lead to budget deficit. To fix such problem, we leverage the underlying idea of competition enhancement (Chu and Shen, 2006, 2007, 2008; Chu, 2009; Roughgarden et al., 2020). To ensure budget balance, incentive compatibility, and high efficiency in double auctions, Chu and Shen (2006) propose a multistage design approach. Based on this methodology, they develop the agent competition double auction mechanism for the single-unit exchange setting with transaction costs. In contrast to the trade reduction approach, the multistage design framework enables the construction of mechanisms

suited to more complex exchange environments (Chu and Shen, 2007). Chu and Shen (2008) introduce a class of truthful mechanisms for the bundle/single-unit context with transaction costs. Chu (2009) presents the “padding” method and designed truthful double auctions for the bundle/multi-unit environment, both with and without the single-minded assumption. In a one-sided market, Roughgarden et al. (2020) define a notion of setting a limit on heterogeneous supply for one-sided matching markets to achieve higher revenue. In their work, *limited supply* refers to constraining the quantity of tradable items. Indeed, the concept of limited supply serves as a form of competition enhancement (Chu and Shen, 2006, 2007, 2008; Chu, 2009), as scarcity can stimulate higher revenue outcome. This paper uses the concept of limited supply to bilateral environments, where *limited supply* can equivalently be interpreted as restricting the number of winners in the swap scenario. We further consider two possibilities of improvements. First, since the meta-items of two agents may be remarkably different, one’s valuation on the ownership of other items can be *negative*. Second, in addition to the swap of meta-items, each agent has the option to *consign* the usage right of their digital goods to a third-party platform for profit.

3 Preliminary survey and examples

3.1 Empirical observation

In the study, an online survey (via the Tencent Questionnaire platform) using a convenience sampling approach was conducted. Before the formal survey, we invited 36 colleagues and students to take a pilot study to identify and correct any mistakes in the measurement instrument design and estimate the time required to complete the questionnaire. Our formal survey includes two types of participants. The first type comprises employees from a leading cultural asset and equity swap organization in China, with 79 valid questionnaires collected. Second, we invited students (including undergraduates, MBA candidates, and Ph.D. students) to distribute the questionnaire among their relatives and friends, resulting in 93 valid responses.

Purchase and collection value, willingness to swap, impact on market liquidity, and auction model selection are key metrics used to evaluate the acceptance and future prospects of digital collectible auctions. Our main observations are summarized in Table 1. While both professional institutional employees and general potential consumers demonstrate overall recognition and optimism toward digital collectible auctions, their survey responses reveal statistically significant differences in key aspects. Specifically, 91.14% of institutional professionals consider digital collectibles to possess significant or moderate purchase/collection value, while the positive response rate among potential consumer groups stands at 75.27%. Similarly, 96.20% of institutional professionals express willingness to participate in digital collectible swaps, compared to 79.57% among consumer respondents. Moreover, 96.21% of professionals believe the auctions positively impact the market (versus 73.12% of consumers), with 93.67% willing to participate (versus 69.89%). This discrepancy may be attributed to institutional employees’ deeper understanding of digital collectibles. Regarding auction model preferences, both professionals and consumers demonstrate greater acceptance of B2B and C2C models compared to B2C. Notably, however, the C2C auction model currently lacks full legal authorization in

China.

Table 1. Survey results: enterprise vs. consumer preferences in digital assets trading

Key Metrics		Enterprise Acceptance	Consumer Acceptance
Purchase & Collection Value		91.14%	75.27%
Willingness to Swap		96.20%	79.57%
Impact of Auctions on Market Liquidity		96.21%	73.12%
Participation Willingness		93.67%	69.89%
Auction Model	B2B	87.34%	72.04%
Preference	B2C	83.54%	68.82%
	C2C	89.87%	73.12%

3.2 Simple examples

Four straightforward examples are now illustrated for the application of limited supply and platform escrow method to achieve budget balance (BB). In this section, since the auction mechanisms ensure agents telling truth, their XOR bids can be treated as true valuations for ease of notation.

3.2.1 The VCG auction

In our swap setting, an agent’s valuations represent his true valuation of another agent’s digital collectible ownership. Agents submit their XOR bids to the third-party platform (auctioneer). The example features three metaverse participants (A, B, and C), and their valuations (XOR bids) for corresponding meta-items are detailed in Table 2. For instance, participant A values his own meta-item at $v_{AA} = 0$, and values $v_{AB} = 5$ for agent B’s meta-item and $v_{AC} = 9$ for agent C’s meta-item. The interpretation of the remaining rows is analogous.

Especially in the table, when $v_{CB} = -1$, it indicates that if agent C wants to acquire the ownership of agent B’s meta-item but lose his own meta-item’s ownership, the value attributed to agent B’s meta-item (i.e., the true assessment) is -1 , implying a “real cost” of 1. It is suggested that from agent C’s standpoint, the desire for agent B’s meta-item exists, yet the perceived value of the meta-item held by agent B is smaller relative to his own meta-item. Therefore, the negative value in the table may exhibit.

Let us initially examine a conventional VCG auction. It is evident that optimal social welfare is realized when player A receives player C’s meta-item, player B receives player A’s meta-item, and player C receives player B’s meta-item. In this allocation, the social welfare $\pi(N)$ is

$$\pi(N) = v_{AC} + v_{BA} + v_{CB} = 9 + 3 + (-1) = 11 \quad (1)$$

where N denotes the sets of players.

Truthful bidding in a VCG auction is achieved by assessing each agent’s payment based on the external impact they have on other participants. The payments are calculated by first excluding one player from the auction and then measuring the difference in the payoffs of the

Table 2. An example of valuations for digital asset swap

	A	B	C
A	0	5	9
B	3	0	4
C	1	-1	0

Table 3. An example of valuations for digital asset swap and escrow

Scheme		A	B	C
Swap	A	0	5	9
	B	3	0	4
	C	1	-1	0
Escrow	Platform Value (b_i)	9	6	3
	Agent Cost (s_i)	1	1	1

remaining agents. Considering agent A as an example, first removing agent A's row and column from Table 2, then it can be observed that the maximal outcome is achieved through the swap of meta-items between agents B and C. The corresponding social welfare is $v_{BC} + v_{CB} = 4 + (-1) = 3$. The agent A's payment is calculated by

$$p_{AC} = v_{AC} - [\pi(N) - \pi(N \setminus \{A\})] = 9 - (11 - 3) = 1 \quad (2)$$

where $\pi(N \setminus \{A\})$ denotes the welfare in the absence of agent A from the auction. The other two agents' respective payments, p_{BA} and p_{CB} , are figured out in a comparable manner

$$p_{BA} = v_{BA} - [\pi(N) - \pi(N \setminus \{B\})] = 3 - (11 - 10) = 2 \quad (3)$$

$$p_{CB} = v_{CB} - [\pi(N) - \pi(N \setminus \{C\})] = -1 - (11 - 8) = -4 \quad (4)$$

While the meta-item distribution to agents optimizes social welfare and is rational for each player, the overall payoff of the platform (auctioneer), which is $1 + 2 + (-4) = -1$, implies that the platform (also named platform D) is operating with a deficit.

Panel (a) of Fig. 2 illustrates the allocation of meta-items and the related payments in a conventional VCG auction. The platform facilitates the trade of meta-items as a third-party entity. The money flows are directed between the agents and the platform, with the ownerships if meta-items being swapped directly among the agents. In the traditional VCG case, all three meta-items are involved in the trade. Agents A and B pay the platform, while Agent C receives a payment to ensure truthful bidding. Consequently, the platform operates at a deficit, suggesting that it would not be sustainable without public subsidies.

3.2.2 The VCG auction with escrow method

The next question is whether introducing a platform escrow mechanism could help achieve budget balance (BB). If an agent chooses to escrow his meta-item with the platform, the actual transaction only involves the transfer of the item's usage right rather than ownership. Under

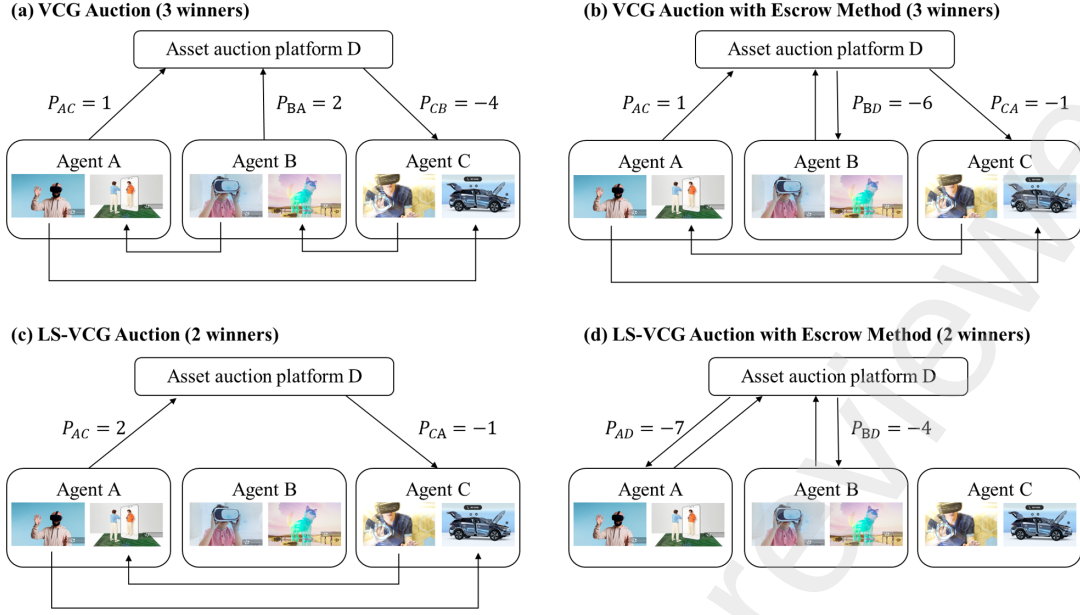


Fig. 2. LS-VCG auction vs. LS-VCG auction with escrow method

the escrow scheme, agents submit their bids (reported costs) to the platform. For the purpose of analysis, we suppose the platform (auctioneer) reveals its real valuations for the rights to use meta-items. Interestingly, in Section 5.2.5 we demonstrate that truthful telling is almost an optimal strategy for maximizing the platform's revenue.

As shown in Table 3, we set that each agent values the usage right of his own meta-item at 1 (cost), denoted by s_i . The platform's valuation for the usage right of each agent's item (b_i) is also specified in the fourth row in Table 3. Therefore, with the introduction of platform escrow in the auction, the meta-items allocation is changed. When agent A and agent C swap the ownerships of their meta-items, and agent B escrow the usage right of his meta-item with the platform, then the highest welfare is realized. Given the context, the social welfare $\pi(N)$ can be expressed as

$$\pi(N) = v_{AC} + v_{BD} + v_{CA} = 9 + (-1 + 6) + 1 = 15 \quad (5)$$

Here, N represents the collective set of agents and the platform (D). Similarly to the procedure in Section 3.2.1, we determine agent A's payment by removing his corresponding rows and columns. Upon agent A's removal, agents B and C consign the usage rights of their meta-items to the platform. This results in a total welfare calculation of $v_{BD} + v_{CD} = (-1 + 6) + (-1 + 3) = 7$. Agent A's payment p_{AC} is derived as follows

$$p_{AC} = v_{AC} - [\pi(N) - \pi(N \setminus \{A\})] = 9 - (15 - 7) = 1 \quad (6)$$

in which $\pi(N)$ represents the total social welfare with all players involved, and $\pi(N \setminus \{A\})$ means the total social welfare if player A is not included. The payment for player B and player C, p_{BD} and p_{CA} are found in the same way

$$p_{BD} = v_{BD} - [\pi(N) - \pi(N \setminus \{B\})] = -1 - (15 - 10) = -6 \quad (7)$$

$$p_{CA} = v_{CA} - [\pi(N) - \pi(N \setminus \{C\})] = 1 - (15 - 13) = -1 \quad (8)$$

We now see that, once we introduce the platform escrow method, total welfare increases from 11 to 15, and the platform's profits increases from -1 to $1 + (-6 + 6) + (-1) = 0$. Hence, the platform neither loses nor gains money, which means that the escrow method contributes to achieving BB.

Panel (b) of Fig. 2 presents a summary of outcomes from the VCG auction with an escrow method. Under the scenario, players A and C swap their meta-items, accompanied by respective payments to the platform. Agent B entrusts his meta-item with the platform and the cost is -1 , while the platform pays him 6. This process remains individually rational for each agent.

3.2.3 The LS-VCG auction

As inspired by Chu and Shen (2006, 2007, 2008), we adopt the concept of *limited supply*, which implies that the maximal count of heterogeneous supply is limited to a certain value and such method can promote the optimal revenue. We consider VCG auction with limited supply (LS-VCG auction) for the example in Table 2. We constrain the maximum number of winners to 2. For the combination of A and C as winners, exchanging their meta-items reaches the highest welfare to $v_{AC} + v_{CA} = 9 + 1 = 10$.

We then calculate the payments for each agent. The payments for agents A and C are

$$p_{AC} = v_{AC} - [\pi(N) - \pi(N \setminus \{A\})] = 9 - (10 - 3) = 2 \quad (9)$$

$$p_{CA} = v_{CA} - [\pi(N) - \pi(N \setminus \{C\})] = 1 - (10 - 8) = -1 \quad (10)$$

When we restrict the count of winning bidders to 2, the aggregate welfare reduces from 11 to 10, but the platform's revenue improves, transitioning from a deficit of -1 to a surplus of $2 + (-1) = 1$. Thus, LS-VCG auction is effective in achieving BB.

Panel (c) of Fig. 2 offers a summary of the outcomes from the LS-VCG auction. Players A and C swap meta-items and pay the platform in this particular scenario. Agent B keeps his own meta-item without having to pay anything. Each agent still finds the process individually reasonable. Additionally, the platform receives a positive total payment.

3.2.4 The LS-VCG auction with escrow method

As demonstrated in the previous example, platform escrow method achieves BB by reducing agents' profits, and meanwhile limited supply does so at the expense of social welfare. Each of the two methods contributes positively to platform's surplus. What would be the result if the platform escrow is employed concurrently with limited supply? We attempt to incorporate the LS-VCG auction with platform escrow (LSE-VCG auction) for the example in Table 3.

We begin by setting the upper limit of winners at 2. When players A and B are selected as the winners, they will consign the usage rights of their meta-items to the platform (D), while

player C retains his own. This configuration yields the maximum welfare, i.e., $v_{AD} + v_{BD} = (-1 + 9) + (-1 + 6) = 13$.

Once again, the player A' payment is calculated by removing player A from the auction, then players B and C choose to escrow their meta-items. The corresponding welfare is $v_{BD} + v_{CD} = (-1 + 6) + (-1 + 3) = 7$, and player A's payment to platform is

$$p_{AD} = v_{AD} - [\pi(N) - \pi(N \setminus \{A\})] = -1 - (13 - 7) = -7 \quad (11)$$

The corresponding payment for agent B, p_{BD} , is computed analogously:

$$p_{BD} = v_{BD} - [\pi(N) - \pi(N \setminus \{B\})] = -1 - (13 - 10) = -4 \quad (12)$$

Now, we observe that LSE-VCG auction increases the platform's profits from 0 to $(9 - 7) + (6 - 4) = 4$, which is higher than the outcomes obtained by employing either limited supply alone or the sole escrow method. Panel (d) of Fig. 2 provides the detailed outcomes of LSE-VCG auction.

The following sections will make these ideas more precise and explore the broader impacts of limited supply and escrow method using mathematical analysis and simulations.

4 The LS-VCG auction with platform escrow

4.1 Model

Consider a digital asset trading platform (e.g., Foundation NFT company, organization X) that includes multiple agents. Within the network, each agent possesses a unique meta-item and has the intention to acquire a different one³. Moreover, such meta-item swap can improve market liquidity and satisfy social demand⁴. Indeed, there are some leading NFT marketplaces for digital collectibles trading and multilateral matching, such as Rarible, OpenSea, and Crypto.com⁵. Also, agents can opt to consign the usage rights of their meta-items to the platform for a certain cost, thereby receiving payments from the platform⁶. This is reasonable because by analyzing view counts platforms (e.g., Spotify, YouTube, Instagram) can assess the potential market value of digital collectibles and determine acquisition interest.

Define v_{ij} as the valuation that player i attributes to the meta-item held by player j , for all i, j within the agents set N . Each agent's valuation remains private, known exclusively to agent i . We aim to design a scheme that ensures agents reveal their truthful valuations. When agent i does not receive a different meta-item, his reservation value is null, denoted as $v_{ii} = 0$. For brevity of analysis, we assume that the platform tells the truth during the bidding process.

The platform gathers basic information regarding agents' meta-items and conducts the sharing arrangements. This functions within a barter swap framework, where an agent acquires

³As in our questionnaire statistics, 96.20% of institutional professionals express willingness to participate in digital collectible swaps, compared to 79.57 % among consumer respondents.

⁴The argument is also supported by our empirical observations, see Q11 in questionnaire statistics.

⁵<https://coinpaprika.com/news/top-nft-marketplaces-for-digital-collectibles/>

⁶The fundamental concept aligns with the new auction mechanism that enhances expected revenue through augmented demand (Chu and Wu, 2023)..

a meta-item only if his own meta-item is allocated to another agent. Otherwise, the agent may choose either to retain his own meta-item or to entrust its usage right to the platform.

The platform tries to maximize social welfare through the implementation of a combinatorial auction, where multiple units (referred to as meta-items) are simultaneously auctioned. Here, the platform assumes the role of auctioneer, and the agents take on the role of bidders. Bidders can submit an XOR bid that includes several atomic (indivisible) bids. Each atomic bid contains the desired meta-item and the corresponding bid value. Subsequently, for every XOR bid, at most an atomic bid is chosen at one time. The winners in the auction are the players who are allocated a distinct meta-item from the platform.

As inspired by Chu and Shen (2006, 2007, 2008), we generalize the notion of *limited supply* in the presence of platform escrow, which essentially limits the maximum number of winning agents. In fact, the underlying idea is consistent with the new auction theory upon promoting expected revenue with augmented demand and limited supply (Chu and Shen, 2006, 2007, 2008; Chu, 2009; Roughgarden et al., 2020; Chu and Wu, 2023). The concept of limited supply is detailed in the following definition.

Definition 1 (*Limited Supply*). Consider N as the group of participants involved in the auction, where $|N|$ indicates the count of agents within the set. Let G represents the set of actual transactions, and $|G| \leq t$, where t is defined as the number of limited supply.

The number of actual transactions $|G|$, counts two mutual transactions between agents or a single consignment to the platform. We now proceed to formally define the winner determination model:

$$\pi(N) = \max \sum_{i \in N} \sum_{j \in N} x_{ij} v'_{ij} + \sum_{i \in N} y_i (b_i - s'_i) \quad (13)$$

$$\sum_{j \in N} x_{ij} + y_i = 1, \forall i \in N \quad (14)$$

$$\sum_{i \in N} x_{ij} + y_i = 1, \forall j \in N \quad (15)$$

$$\sum_{i \in N} x_{ii} \geq |N| - t \quad (16)$$

$$x_{ij} \leq X_{ij}, \forall i, j \in N \quad (17)$$

$$x_{ij} \in \{0, 1\}, \forall i, j \in N \quad (18)$$

$$y_i \in \{0, 1\}, \forall i \in N \quad (19)$$

Here, v'_{ij} represents the valuation (bid) stated by agent i for agent j 's meta-item; b_i stands for the platform's assessment of the usage right of the meta-item owned by agent i , while s'_i refers to the cost (bid) reported by the agent i . Note that, in our auction mechanisms we guarantee incentive compatibility. Thus, we generally do not distinguish real valuations v_{ij} and s_i with bids v'_{ij} and s'_i throughout this paper. For ease of exposition, we also assume the platform reports his true valuations for the usage rights of meta-items (b_i)⁷.

⁷Note that, in Section 5.2.5 we demonstrate that truthful telling is almost an optimal strategy for maximizing the platform's revenue.

Moreover, x_{ij} is a binary variable signifying the allocation of the meta-item, which takes the value 1 when player i obtains player j 's meta-item, and 0 otherwise (Eq.18). Similarly, y_i is a binary variable indicating whether agent i chooses platform escrow. When $y_i = 1$, it signifies that agent i entrusts the usage right of his meta-item to the platform and is excluded from participating in other meta-item swap; otherwise, y_i takes a value of 0 (Eq.19).

The initial step in the optimization algorithm involves defining a limited supply number t , which dictates the upper limit on the count of winners within the auction (i.e., $|G| \leq t$). This is followed by the identification of the subset G of players N that meets the quantity constraint of t and maximizes social welfare. Further details on determining the optimal scale will be provided subsequently.

The maximization problem ensures that each agent can be allocated to at most one meta-item's ownership, and if an agent chooses to entrust the usage right of his meta-item to the platform, he will not be allocated a meta-item (Eq.14). Simultaneously, each meta-item's ownership can be obtained by at most one agent; if a meta-item is entrusted to the platform, it cannot be obtained by the agents (Eq.15). We employ Eq.16 for limited supply, which specifies that a minimum of $|N| - t$ agents should be assigned their own meta-items with their reserve value of 0. Eq.17 guarantees the swap of meta-items. Here, X_{ij} in Eq.17 equals to 0 when $i \neq j$ and $v_{ij} = 0$, and 1 otherwise. Therefore, if an agent's reservation utility for a meta-item other than their own is zero, the algorithm will avoid allocating that meta-item to the agent. Since that $|N| - t$ players will receive a zero predetermined reservation value, the social welfare from this optimization is the peak obtainable with any subset of t players in N . Through out this section, the notations is presented in Table 4.

We now outline the procedure of the VCG auction with limited supply and platform escrow (called *LSE-VCG auction*). Initially, the platform collects detailed information regarding all the $|N|$ meta-items and communicates this to all agents. Subsequently, each agent presents his assessed valuations v_{ij} for all meta-items. With the platform providing its valuations b_i , each agent reports the consignment costs s_i concurrently. As mentioned earlier, the platform is assumed to tell the truth (b_i) in the auction. The platform applies a maximization approach as presented in Eq.13 to identify the group of winners G , taking into account the limited supply number t . Ultimately, a winner who swaps or sells his meta-item is required to make a payment calculated as

$$p_{ij} = v_{ij} - [\pi(N) - \pi(N \setminus \{i\})] \quad (20)$$

$$p_i = -s_i - [\pi(N) - \pi(N \setminus \{i\})] \quad (21)$$

If the winner consigns the usage right of his meta-item with the platform, the payment will be made as p_i in Eq.21. Note that p_i is negative, implying that the winner receives the payment $-p_i$ from the platform. Clearly, $\pi(N \setminus \{i\})$ represents the social welfare in Eq.13 if the bids of agent i (along with his associated meta-item) were excluded. Therefore, $\pi(N) - \pi(N \setminus \{i\})$ signifies the externality imposed by agent i on all other participates, and thus represents the profits of agent i . It is naturally assumed that ties are resolved arbitrarily.

Accordingly, when an agent i opts to consign the usage right of his own meta-items to the

Table 4. Notation list

Decision Variables	
x_{ij}	Binary variable which equals 1 if agent $i \in N$ obtains the ownership of agent j 's ($j \in N$) meta-item in the auctions
y_i	Binary variable which equals 1 if agent $i \in N$ entrust the usage right of his meta-item to the platform in the auctions
Parameters	
N	The set of agents in the auction
t	The number of limited supply
G	The set of actual transactions in the auction
v_{ij}	The true valuation of agent i ($i \in N$) for the ownership of agent j 's ($j \in N$) meta-item
v'_{ij}	The bidding valuation of agent i ($i \in N$) for the ownership of agent j 's ($j \in N$) meta-item
s_i	The real cost of agent i 's ($i \in N$) meta-item usage right
s'_i	The bidding cost of agent i 's ($i \in N$) meta-item usage right
b_i	The platform's valuation for the usage right of agent i 's ($i \in N$) meta-item
p_{ij}	The payment that a winner i ($i \in N$) who sells the ownership of his meta-item to the buyer j ($j \in N$) should make
p_i	The payment that a winner i ($i \in N$) who consigns the usage right of his meta-item with the platform should make
$u_{e,i}$	The utility that the platform received when an agent i ($i \in N$) consigns the usage right of his meta-item with platform
$\pi(N)$	The total social welfare when all the agent in the auction bids truthfully

platform, the platform's utility is as follows:

$$u_{e,i} = b_i - s_i - [\pi(N) - \pi(N \setminus \{i\})] \quad (22)$$

The subscript e in $u_{e,i}$ denotes the escrow with the platform. This mirrors the strategy employed in traditional VCG auctions in incentivizing truthful telling. However, in this paper we incorporate both limited supply and barter requirements, along with platform escrow. As commonly practiced in auction design research (Bikhchandani and Ostroy, 2006; Xu and Huang, 2017), this paper will adopt the Bayesian Nash Equilibrium notion to demonstrate incentive compatibility within our LSE-VCG auction.

Proposition 1 (*Incentive Compatibility*). *Truthful reporting is a Bayesian Nash Equilibrium in the LSE-VCG auction.*

Proof. Assume that all players truthfully submitted their bids. Suppose player $i \in N$ submits his true valuation v_{ij} for player j 's meta-item, or s_i for his own meta-item. When he is swapped with the other agent, then he is supposed to make payment $p_{ij} = v_{ij} - [\pi(N) - \pi(N \setminus \{i\})]$; when he chooses to escrow with the platform, then his payment will be $p_i = -s_i - [\pi(N) - \pi(N \setminus \{i\})]$. If agent i bids $v'_{ij} \neq v_{ij}$ or $s'_i \neq s_i$, the results will be changed, which can be discussed in four scenarios.

Scenario 1: Agent i who originally swaps meta-item with agent j , may be allocated agent k 's meta-item if he submits valuation $v'_{ij} \neq v_{ij}$. The related payment in this case is $p'_{ik} = v'_{ik} - [\pi'(N) - \pi(N \setminus \{i\})]$, where $\pi'(N)$ denotes the welfare resulting from the allocation, calculated as follows:

$$\pi'(N) = v'_{ik} + \sum_{l \in N' \setminus \{i\}} \sum_{j \in N' \setminus \{k\}} x'_{lj} v_{lj} + \sum_{m \in N' \setminus \{i, k\}} y'_m (b_m - s_m) \quad (23)$$

Here, x'_{lj} and y'_m represent the optimal solution to $\pi'(N)$, N' is the set of resulting winning players, l is any winning player other than player i , and m is any player who consigns their meta-items to the platform other than agents i, k . For player i who benefits from submitting the false bid, the following condition must be met:

$$\pi(N) - \pi(N \setminus \{i\}) < v_{ik} - p'_{ik} \quad (24)$$

Substituting for p'_{ik} yields

$$\pi(N) < v_{ik} + \sum_{l \in N' \setminus \{i\}} \sum_{j \in N' \setminus \{k\}} x'_{lj} v_{lj} + \sum_{m \in N' \setminus \{i, k\}} y'_m (b_m - s_m) \quad (25)$$

The inequality clearly challenges the assumption that $\pi(N)$ signifies an optimal allocation of meta-items.

Scenario 2: Agent i who originally swaps meta-item with agent j , may consign his meta-item with the platform if he submits valuation $s'_i \neq s_i$. The related payment in this case is $p'_i = -s'_i - [\pi'(N) - \pi(N \setminus \{i\})]$, where $\pi'(N)$ denotes the welfare resulting from the allocation, calculated as follows:

$$\pi'(N) = b_i - s'_i + \sum_{l \in N' \setminus \{i\}} \sum_{j \in N' \setminus \{i\}} x'_{lj} v_{lj} + \sum_{l \in N' \setminus \{i\}} y'_l (b_l - s_l) \quad (26)$$

Here, x'_{lj} and y'_l represent the optimal solution to $\pi'(N)$, N' is the group of resulting winning players, l is any winning agent other than agent i . The following condition must be met for agent i who benefits from submitting the false bid:

$$\pi(N) < \pi'(N) + s'_i - s_i \quad (27)$$

The inequality also contradicts the optimal allocation premise of $\pi(N)$.

Scenario 3: Agent i who originally consigns meta-item with the platform may submit an unreal cost $s'_i \neq s_i$. Then the payment will be $p'_i = -s'_i - [\pi'(N) - \pi(N \setminus \{i\})]$, and the corresponding welfare is similar to Eq.26 in scenario 2. For agent i who benefits from submitting the false bid, the following condition must be met:

$$p'_i < -s_i - [\pi(N) - \pi(N \setminus \{i\})] \quad (28)$$

Substituting for p'_i yields

$$\pi(N) < b_i - s_i + \sum_{l \in N' \setminus \{i\}} \sum_{j \in N' \setminus \{i\}} x'_{lj} v_{lj} + \sum_{l \in N' \setminus \{i\}} y'_l (b_l - s_l) \quad (29)$$

The inequality also contradicts the premise that $\pi(N)$ implies an optimal assignment of meta-items.

Scenario 4: Agent i who originally consigns meta-item with the platform may swap meta-item with agent k if he submits valuation $v'_{ij} \neq v_{ij}$. Then the payment will be $p'_{ik} = v'_{ik} - [\pi'(N) - \pi(N \setminus \{i\})]$, and the corresponding welfare is similar to Eq.23 in Scenario 1. And the following condition must be met for agent i :

$$\pi(N) < \pi'(N) + v_{ik} - v'_{ik} \quad (30)$$

This inequality also contradicts the notion that $\pi(N)$ signifies the best possible allocation.

In each of the four scenarios, the argument holds true for any participant in N . Thus, these scenarios confirms the incentive compatibility. ■

Moving forward, there is no need for differentiation between declared valuations and actual valuations. In a barter environment, the feature of truthful telling can simplify bidding strategies to attract customers. This LSE-VCG auction ensures the best outcome when agents reveal their true values, thereby enhancing the efficiency and feasibility. Our focus shifts to demonstrating the individual rationality of the LSE-VCG auction.

Proposition 2 (*Participation Rationality*). *The LSE-VCG auction realizes participation rationality.*

Proof. Clearly, the profits of each agent could be expressed as $\pi(N) - \pi(N \setminus \{i\})$. Specifically, When $\pi(N) - \pi(N \setminus \{i\}) = 0$, agent i remains indifferent towards participation. Conversely, in cases where $\pi(N) - \pi(N \setminus \{i\}) > 0$, agent i experiences a strictly superior outcome, compelling he chooses to participate. Regardless of the two scenarios, agent i opts to participate, thus confirming the individual rationality of the LSE-VCG auction. ■

4.2 A special case: the VCG auction with limited supply

4.2.1 The LS-VCG model

In general, the revenue in an auction can be promoted by competition, which is usually intensified via limiting supply or increasing demand. The VCG auction model incorporating both *limited supply* and platform escrow (LSE-VCG auction) has been established in Section 4.1. In our benchmark model, the sole scheme of limited supply is immediately derivable through elimination of the platform escrow mechanism. The VCG auction with *limited supply* (LS-VCG auction) constrains the count of winners involved in the swap in advance, ensuring that the allocation process remains incentive compatible and individual rational. By limiting the number of winners in the auction, the LS-VCG mechanism achieves budget balance, a property that the traditional VCG auction lacks.

4.2.2 Impossibility theorem

This section establishes the impossibility theorem (Theorem 1) for VCG auctions lacking limited supply in barter environments.

Definition 2 (*Kelso and Crawford, 1982; Gul and Stacchetti, 2000; de Vries et al., 2007*) We say that agents are substitutes if the combined marginal product of a group exceeds the sum of the individual marginal products of the agents within that group, i.e.

$$\pi(N) - \pi(N \setminus N') \geq \sum_{i \in N'} (\pi(N) - \pi(N \setminus \{i\})), \forall N' \subseteq N \quad (31)$$

Lemma 1 *If the number of swaps is $|N|$ that achieves maximal social welfare $\pi(N)$, then the substitute condition does not hold.*

Proof. We argue by induction on $|N|$. If $|N| = 2$, let $N' = N$, then the left-side of Eq.31 equals to $\pi(N)$, but the right-side of Eq.31 equals to $2\pi(N)$. Thus, the result holds when $|N| = 2$.

We next consider the case with $|N| = 3$ and $N' = N$. The left-side of Eq.31 equals to $\pi(N)$. Then, we try to make the right-side of Eq.31 as small as possible; that is, $\pi(N) - \pi(N \setminus \{i\}) < \pi(N)$ for any i . For simplicity, we say $N = \{A, B, C\}$, and the value vector $v = \{v_{AB} = a, v_{BA} = a', v_{BC} = b, v_{CB} = b', v_{CA} = c, v_{AC} = c'\}$. So, the right-side of Eq.31 equals to $\pi(N) - \pi(N \setminus \{A\}) + \pi(N) - \pi(N \setminus \{B\}) + \pi(N) - \pi(N \setminus \{C\})$. Suppose the substitute condition holds. We obtain that $\pi(N) \geq \pi(N) - \pi(N \setminus \{A\}) + \pi(N) - \pi(N \setminus \{B\}) + \pi(N) - \pi(N \setminus \{C\})$; that is, $\pi(N) = a + b + c \leq a' + b' + c'$, which causes a contradiction. Note that, ties are broken arbitrarily, so $\pi(N)$ is the unique optimal solution. Thus, the result holds when $|N| = 3$.

Similarly, in the case with $|N| = 4$ and $N' = N = \{A, B, C, D\}$, if the substitute condition holds, then we have that

$$\pi(\{A, B, C\}) + \pi(\{A, B, D\}) + \pi(\{A, C, D\}) + \pi(\{B, C, D\}) \geq 3\pi(N) \quad (32)$$

Observe that each winner is covered by at most three swap cycles. Then, by a simple construction, the left-side of Eq.32 contains at most three feasible solutions in the same time. Clearly, the constructed feasible solutions cannot be the same and must be less than π . So, the sum of the constructed feasible solutions is strictly less than $3\pi(N)$, which causes a contradiction. Thus, the result holds when $|N| = 4$.

Finally, in the case with $|N| = k$ and $N' = N$, if the substitute condition holds, then we obtain that

$$\sum_{i \in N} \pi(N \setminus \{i\}) \geq (h-1)\pi(N) \quad (33)$$

Likewise, the left-side of Eq.33 contains at most $h-1$ feasible solutions, implying a contradiction. Thus, the result holds when $|N| = h$.

From the above argument, Lemma 1 has been proved. ■

Lemma 2 *If the number of swaps is $|N| - 1$ that achieves maximal social welfare $\pi(N)$, then the substitute condition does not hold.*

Proof. Suppose $|N| = h$ and let I denote the set of winners. If the substitute condition holds, then we have

$$\sum_{i \in I} \pi(N \setminus \{i\}) \geq (h - 2)\pi(N) \quad (34)$$

Let j be the (unique) loser. Observe that we can arbitrarily add new arcs ($\tilde{v}_{ij}$ or $\tilde{v}_{ji}$) to agent j and/or improve some values of agent j so as to guarantee that all agents are winners and $\pi(I) = \pi(I \cup \{j\})$. For example, we can add new arcs $\tilde{v}_{ij}$ and $\tilde{v}_{jk}$, satisfying $\tilde{v}_{ij} + \tilde{v}_{jk} = v_{ik}$. If the arcs v_{ij} and v_{jk} have already existed, then we can improve the weights of such arcs so that $\tilde{v}_{ij} + \tilde{v}_{jk} = v_{ik}$. Note that, when only v_{ij} has existed, we can add the arc $\tilde{v}_{jk}$ such that $v_{ij} + \tilde{v}_{jk} = v_{ik}$, where $\tilde{v}_{jk}$ is negative if $v_{ij} > v_{ik}$. By such construction, the left-side of Eq.34 becomes weakly larger since there are more feasible solutions. Interestingly, the right-side of Eq.(34) is unaltered.

So far, we have found two optimal solutions, $\pi(I)$ and $\pi(I \cup \{j\})$. Then, we broke ties and choose $N = I \cup \{j\}$ as the unique set of winners. That is, $\pi(I \cup \{j\})$ is the unique optimal solution though the gap between $\pi(I)$ and $\pi(I \cup \{j\})$ can be completely ignored.

By simultaneously adding π to the left-side and the right-side of Eq.34, we obtain

$$\sum_{i \in I} \pi(N \setminus \{i\}) + \pi(N \setminus \{j\}) \geq (h - 1)\pi(N) \quad (35)$$

The Eq.35 can be rewritten as

$$\sum_{i \in N} \pi(N \setminus \{i\}) \geq (h - 1)\pi(N) \quad (36)$$

According to Lemma 1, we argue that Eq.36 does not hold.

Thus, we have proved Lemma 2. ■

Theorem 1 (Impossibility Theorem) *If the multilateral matching achieves maximal social welfare π , then the substitute condition does not hold.*

Proof. By a similar construction stated in the proof of Lemma 2, we can find two optimal solutions, $\pi(I)$ and $\pi(I \cup (N \setminus I))$, where $N \setminus I$ is the set of losers in the solution $\pi(I)$. That is, we can find an optimal solution by adding arcs to the set of losers and/or improving the values of the losers so that the set of winners is N .

Likewise, if the substitute condition holds, then we have

$$\sum_{i \in I} \pi(N \setminus \{i\}) + \sum_{j \in N \setminus I} \pi(N \setminus \{j\}) \geq (h - 1)\pi(N) \quad (37)$$

Further, we get

$$\sum_{i \in N} \pi(N \setminus \{i\}) \geq (h - 1)\pi(N) \quad (38)$$

Clearly, the Eq.38 contradicts Lemma 1. Thus, we have proved Theorem 1. ■

By a similar argument in Theorem 3 of Xu et al. (2018), the VCG-like auction in a swap context is budget-balanced if buyers are substitutes. However, our impossibility theorem does

not implies that such auction results in budget deficit if maximal social welfare is achieved with multilateral matching. The subsection finds the importance of limited supply, which improves platform's surplus in some conditions.

4.2.3 Importance of limited supply

Theorem 2 *The platform's surplus is weakly larger under $t = |N| - 1$ than that under $t = |N|$.*

Proof. Observe that if the number of swaps under maximal social welfare is strictly less than $|N|$, then the platform's surplus is identical whether $t = |N| - 1$ or $t = |N|$.

Suppose the number of swaps under maximal social welfare is $|N|$ and the optimal solution is unique (ties are broken arbitrarily). Then, the platform's surplus under $t = |N|$ is given by

$$s(t = |N|) = \sum_{i \in N} \pi(N \setminus \{i\}) - (|N| - 1)\pi(N, t = |N|) \quad (39)$$

Now we consider the case with $t = |N| - 1$. In this case, the platform's surplus is given by

$$s(t = |N| - 1) = \sum_{i \in I'} \pi(N \setminus \{i\}) - (|I'| - 1)\pi(N, t = |N| - 1) \quad (40)$$

where I' is the set of winners.

Clearly, the Eq.39 can be rewritten as

$$s(t = |N|) = \sum_{i \in I'} \pi(N \setminus \{i\}) + \sum_{j \in N \setminus I'} \pi(N \setminus \{j\}) - (|N| - 1)\pi(N, t = |N|) \quad (41)$$

Also, notice that for any $j \in N \setminus I'$, we have

$$\pi(N \setminus \{j\}) = \pi(N, t = |N| - 1) \quad (42)$$

From Eq.39, Eq.41 and Eq.42, it follows that

$$s(t = |N| - 1) - s(t = |N|) = (|N| - 1) [\pi(N, t = |N|) - \pi(N, t = |N| - 1)] > 0 \quad (43)$$

The inequality above implies that the platform's surplus is strictly larger under $t = |N| - 1$ than that under $t = |N|$ if the number of swaps under maximal social welfare is $|N|$.

Thus, we have proved Theorem 2. ■

Theorem 2 is quite useful because it clearly shows that setting $t = |N| - 1$ is weakly better off for the platform. Given the complexity involved in the analysis, we conduct extensive simulations to further validate the importance of limited supply and platform escrow.

4.2.4 Prediction of limited supply

Given the full bidding information, the *optimal* limited supply number can be calculated to maximize the platform's surplus. Similarly, one can obtain the *maximal* limited supply number

to realize budget balance. However, in LSE-VCG auction the limited supply number is predetermined. We next use a machine learning method to predict desirable limited supply numbers based on limited information, which is revealed before the auction. The limited information includes the number of agents, the number of atomic bids per XOR bid, and the range of bid value. For simplicity, we focus on the prediction method for LS-VCG auction.

Our predictive framework unfolds in the following moves. Initially, a training dataset is compiled, encompassing variables from multiple auctions, such as the number of agents, bid count and bid value, as well as the *optimal* and *maximal* limited supply numbers, as determined by the LS-VCG model. Subsequently, an appropriate machine learning model is selected, which is then thoroughly trained and validated against the training dataset. Ultimately, a test dataset is applied to evaluate the accuracy and robustness of predictive model. The predictive effect of the model is evaluated by some evaluation metrics.

(1) Model training and data sets The initial step in model training involves amassing a substantial dataset to learn patterns from historical data. Given the sensitive and proprietary nature of auction data, typically accessible only to the auction platform, we generate a training dataset. The rules for generating the training dataset are as follows. The number of agents is set between 20 and 60, with each agent submitting 3, 6, or 9 (atomic) bids. The value of each atomic bid is calculated as the agent's valuation of another agent's meta-item minus the valuation of their own meta-item. Through this approach, a training set comprising 1200 samples is generated, providing ample data support for model training.

Due to the scarcity of similar research upon auction design, a basic linear regression model is opted for applicability and interpretability. Regarding the selection of independent variables, we only consider information readily available before an auction begins. Such limited information includes the number of agents, the number of atomic bids per XOR bid, and the range of bid value. Two prediction objectives are set: one for maximizing platform revenue and the other for ensuring platform financial balance.

The data generation rules for the test set are analogous to those of the training set, yet with certain distinctions. The number of agents in the test set is randomly generated within the interval $[20, 60]$, and the number of bids per agent falls randomly within $[2, 9]$. The minimum bid value is randomly determined within $[-10, 10]$, while the bid value range is established by a random figure drawn from $[10, 50]$. The maximum bid value is obtained by summing this range with the minimum bid value. Within this defined span, bid values for each bidder are then randomly generated. Although this method differs from the training set generation process, it ensures a broad consistency in data characteristics. The test data set consists of 100 samples.

The entire predictive process is depicted in Fig. 3. The complete information after the auction ends, along with the calculated limited supply numbers, forms the basis for the predictive process. Subsequently, the key data features from the complete data are input into the model to forecast the limited supply number. The trained predictive model serves as a valuable tool for third-party auction platforms. Utilizing this model, the platform can input partial information before an auction starts to anticipate a limited supply number that optimizes his interests. The objective of the forecast is to enhance platform profitability and mitigate the risk of financial

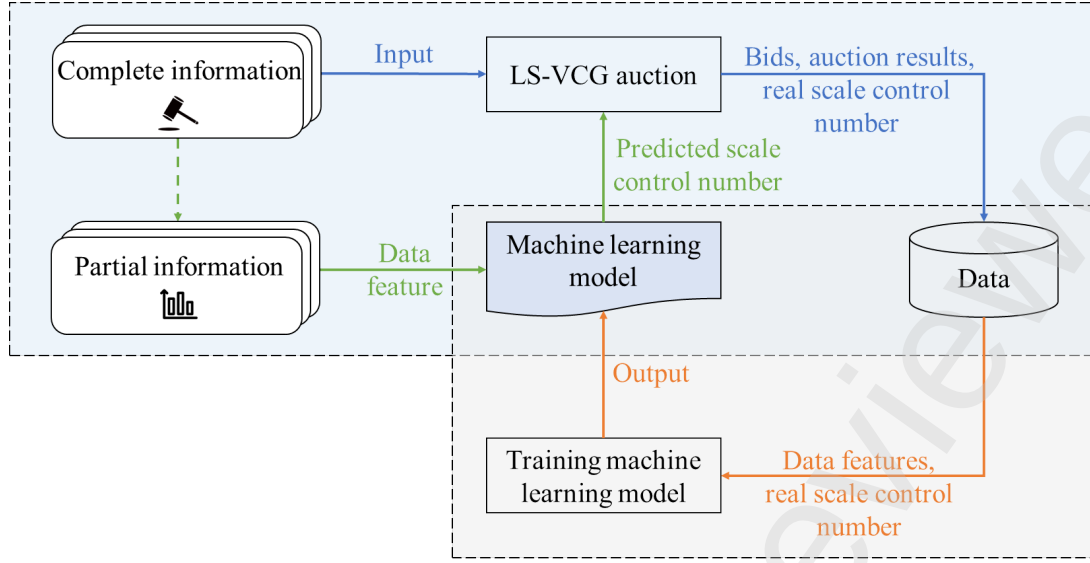


Fig. 3. Predictive process of auction outcomes using incomplete and complete data

losses.

(2) Predictive modeling and result analysis In accordance with the predictive process, the two linear regression models for our forecast objectives are shown as

$$y_{opt} = -0.12x_1 + 0.66x_2 + 1.55x_3 - 3.42 \quad (44)$$

$$y_{max} = -0.08x_1 + 0.92x_2 + 1.30x_3 - 4.76 \quad (45)$$

Here, x_1 , x_2 and x_3 represent the bid value range, the number of bidders, and the number of bids, respectively; y_{opt} indicates the limited supply number t predicted for maximizing platform surplus, while y_{max} denotes the limited supply number t predicted for achieving minimum positive platform surplus. Interestingly, y_{opt} and y_{max} are negatively correlated with the bid value range. As expected, y_{opt} and y_{max} are positively correlated with the number of bidders and the number of bids.

Since high-dimensional models cannot be effectively depicted in two dimensions, relevant evaluation metrics are applied to assess the models' predictive performance. The predicted t 's errors are calculated as

$$e_{opt} = \frac{|\hat{t}_{opt} - t_{opt}|}{t_{opt}} \quad (46)$$

$$e_{max} = \frac{|\hat{t}_{max} - t_{max}|}{t_{max}} \quad (47)$$

In the above equations, $\hat{t}_{opt}$ and $\hat{t}_{max}$ denote the predict t for maximum and minimum positive platform surplus. The real t for the same objective is represented by t_{opt} and t_{max} . The prediction errors for $\hat{t}_{opt}$ and $\hat{t}_{max}$ are indicated by e_{opt} and e_{max} .

Prediction errors for the test data set are plotted in Fig. 4. The scatter plots illustrate prediction errors, with trend lines fitting the error points. The blue areas reflect the scatters' concentration, with larger areas indicating greater dispersion. The subfigure (a) in Fig. 4 reveals

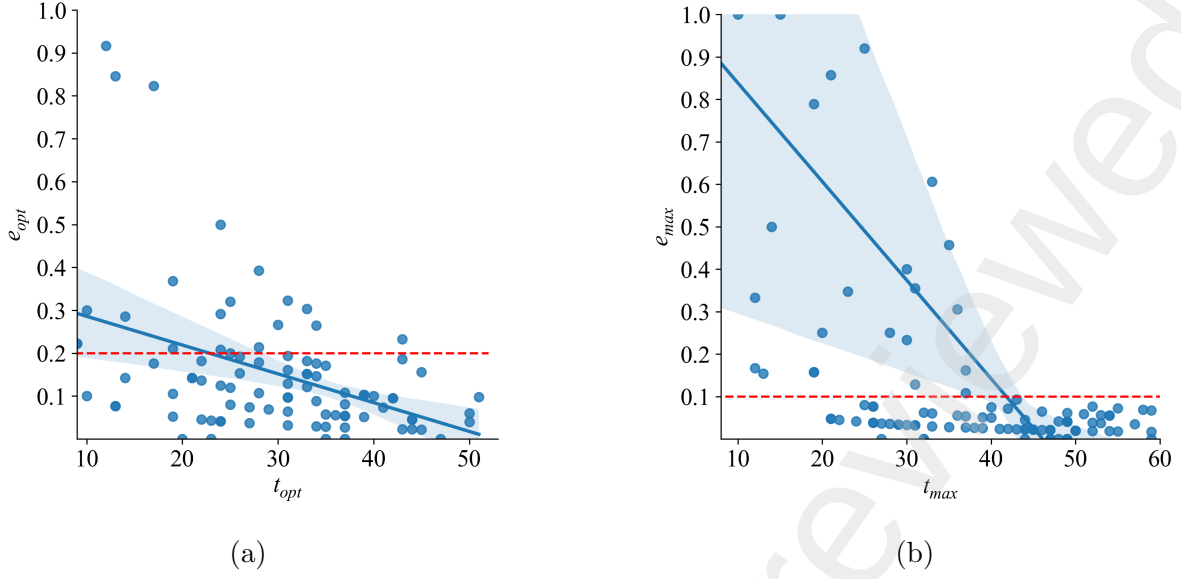


Fig. 4. Errors of predicted t for maximum and minimum positive platform surplus

that e_{opt} are consistently near 20%, and mean error is 15.3%. The subfigure (b) in Fig. 4 shows large prediction errors for corresponding $t_{max} < 20$ with marginal positive revenue. But the errors decrease for $t_{max} > 40$, keeping errors under 10%, and mean error is 21.7%.

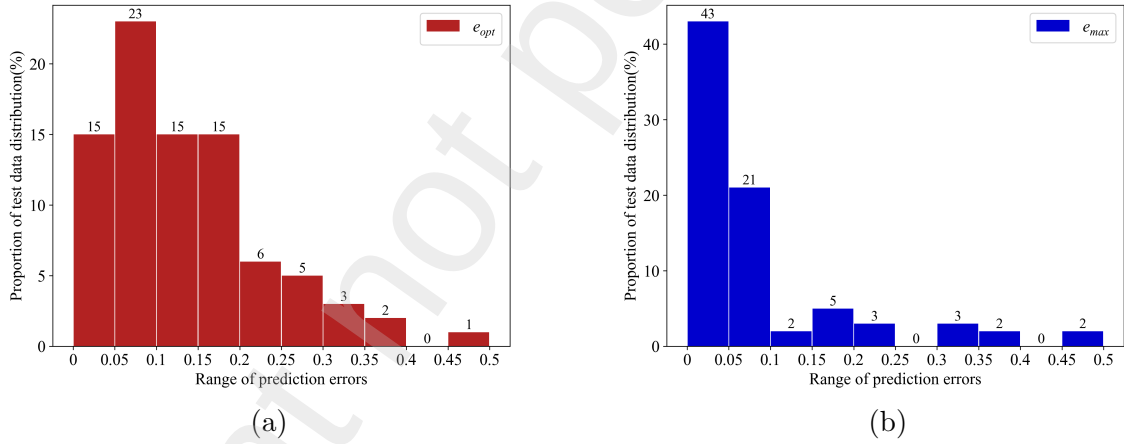


Fig. 5. Distribution of prediction errors for maximum and minimum positive platform surplus

Fig. 5 presents probability histograms of e_{opt} and e_{max} . Table 5 summarizes the proportion of these errors within the overall test dataset. It is clear that the prediction accuracy for maximizing platform revenue is less concentrated compared to the minimum positive revenue scenario. The prediction of minimum positive revenue shows higher concentration but is sensitive to slight variations in t , risking negative outcomes. A relatively conservative platform can use a smaller limited supply number than $\hat{t}_{max}$ to avoid budget deficit. The auction platform could therefore achieve a high surplus using our machine learning method. In the long run, the platform which uses the $\hat{t}_{max}$ can also retain more participants.

Overall, the application of straightforward machine learning models within the auction domain offers the third-party platform insightful reference data. Moving forward, it would be

Table 5. Cumulative distribution table of prediction errors

Predict error range	Portion of test data	
	Maximum PS	Minimum positive PS
$\leq 5\%$	15%	43%
$\leq 10\%$	38%	64%
$\leq 15\%$	53%	66%
$\leq 20\%$	68%	71%

beneficial to explore a broader range of machine learning training methods. Through leveraging real auction data can markedly enhance the models' predictive precision and robustness.

5 Simulations

This paper performs a series of numerical tests to evaluate how well the LS-VCG auction with platform escrow (LSE-VCG auction) is doing. We compare how well the LSE-VCG auction does against the LS-VCG auction evaluating four key metrics: the resulted social welfare, the rate of meta-item swap, the platform's surplus and the agents' profits. We also conduct an examination into the effects of the platform's real valuation of agents' meta-items, whether high or low, on the outcomes of final auctions. Additionally, we explore the influence of the platform's selfish telling and batch size on the auction outcomes.

5.1 Experimental setup

Our test initiates with a meta-item swap market that includes 80 agents⁸. The value each agent places on his own meta-item follows a uniform distribution from 0 to 20. Conversely, the value they assign to another agent's meta-item adheres to a uniform distribution ranging from 10 to 50. **Agents are allowed to place one XOR bid consisting of 3 distinct atomic bids⁹, where the value of each atomic bid is calculated as the agent's valuation of another agent's meta-item ownership minus the valuation of their own meta-item. The cost for consigning the meta-items matches the valuation of the usage rights.** Additionally, the platform's valuation of any agent's meta-item is randomly selected from a uniform distribution from 10 to 50. Since both auctions have been shown to be truthful, bidding prices correspond to real valuations.

The results discussed are based on the average of 20 stochastically produced cases. For the single case, the XOR bids are generated stochastically too. This work develops the model coding and algorithm execution in Python and COPT. Utilizing COPT 5.0, we address the binary programs associated with the LS-VCG and LSE-VCG auctions. All experiments were conducted on a machine equipped with a 12th Gen Intel(R) Core (TM) i7-12700H processor running at 2.30 GHz, along with 16 GB of RAM.

⁸The market size in an auction is usually below 500, as in our questionnaire statistics.

⁹We invited 36 colleagues and students to participate in a pilot study. Using a generative AI model——DALL·E 3, we created unique digital artworks for each participant. Low-resolution previews of all artworks were shared in a WeChat group, where participants could select pieces from others for potential swaps. The resulted selection quantity ranged from 0 to 5 pieces per participant. Based on the results, we set 3 pieces per participant in our benchmark test.

In assessing the comparative performance of the LS-VCG and LSE-VCG auctions, this work evaluates the stability of findings via adjusting quantities of atomic bids (either 3, 6, or 9) and the number of agents (40, 80, or 120).

5.2 Results

We evaluate the LSE-VCG auction's performance across six key variables. Firstly, the platform's (average) surplus, indicated by PS in figures, represents total payoff of the platform. Secondly, the (average) profits of agents, labeled as AP in figures. Thirdly, the (average) social welfare is the combined total of the profits gained by the agents and the surplus earned by the platform, which is marked by SW in figures. The fourth is the (average) swap ratio of meta-items, signified by RE in figures. The fifth variable is the average escrow ratio for the platform, noted as RPE in figures. It should be clarified that RE includes the overall rates of swaps among agents and consignments to the platform (RPE). The sixth metric compares the current social welfare of the LSE-VCG auction to that of the traditional VCG auction without limited supply, represented by SW/Best-VCG-SW in figures.

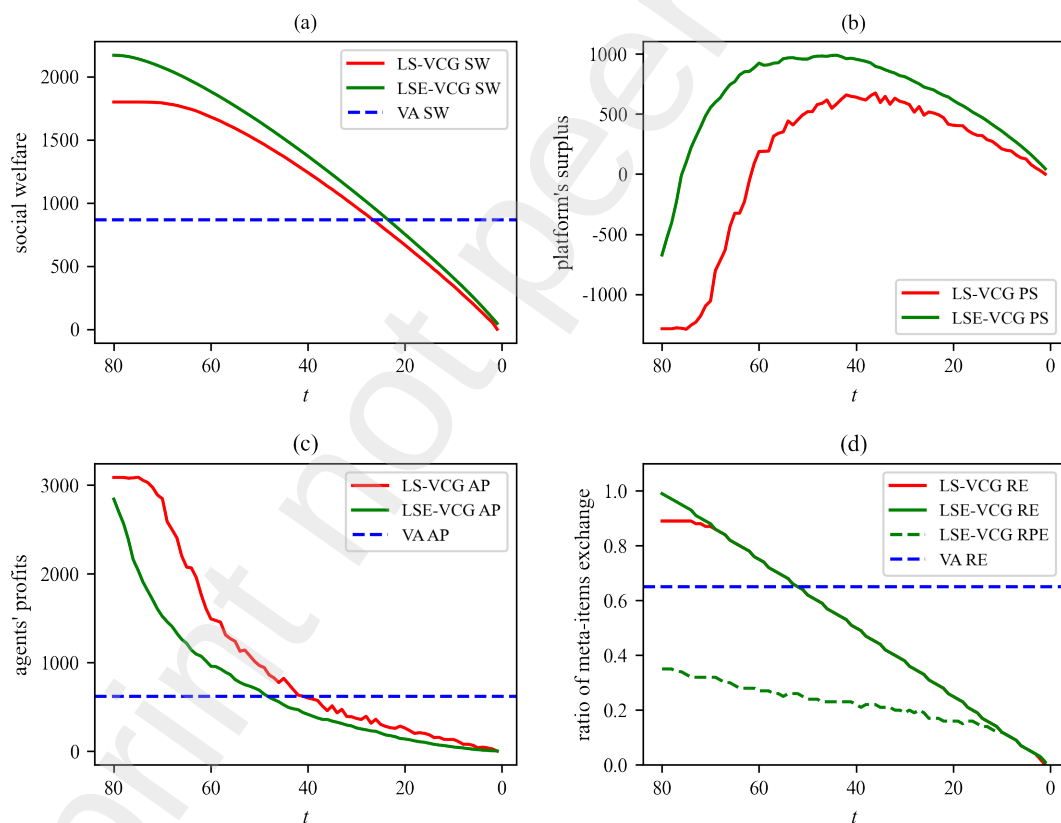


Fig. 6. The effect of platform escrow method

5.2.1 The effect of platform escrow method

Fig. 6 shows how limited supply and the platform escrow method affect each performance metric. To facilitate comparison, we also conducted sequential Vickrey auctions under the same conditions, and we present the performance metrics achieved solely through limited supply. In

the subplots of Fig. 6, the green lines represent LSE-VCG performance with platform escrow, the red lines show the LS-VCG auction results, and the blue lines illustrate the Vickrey auction outcomes.

In the three subplots (a), (c), and (d) of Fig. 6, the blue curves are far below the peaks of the red and green curves, suggesting that the social welfare and agents' profits from the sequential Vickrey auctions are much lower than the best outcomes of LS-VCG auction. According to the Revenue Equivalence Principle (Myerson, 1981; Riley and Samuelson, 1981; McAfee and McMillan, 1987; Maskin et al., 1989), the expected revenue from the Vickrey auction is equivalent to that of any standard auction with risk-neutral and symmetric bidders. Consequently, the optimal experimental results of LS-VCG auction indeed surpass those of the conventional auctions, such as English auction, Dutch auction, the first-price sealed auction and Vickrey auction.

In the three subplots (a), (b), and (d) of Fig. 6, the green curves consistently lie above the red curves or coincide with them, indicating that under equivalent conditions, the performance measures of LSE-VCG auction outperform those of the VCG auction solely employing limited supply. Moreover, the disparity between these two curves narrows as the limited supply number t decreases. With an appropriate choice of t , employing LSE-VCG auction can further enhance social welfare and platform's surplus.

In the subplot (d) of Fig. 6, the upper green curve represents the aggregate real swap ratio under LSE-VCG, while the lower green curve depicts the practical transaction ratio between the platform and agents. It is observed that as the limited supply number t gradually decreases to zero, the two green curves gradually converge. At sufficiently low values of t , agents are more inclined to entrust their meta-items to the platform.

When $t = 75$, LSE-VCG auction has achieved budget balance with $PS = 100$, contrasting with the LS-VCG auction's $PS = -1290.23$ at the same threshold. Upon reduction to $t = 61$, the PS of LS-VCG auction turns negative to positive, with the $PS = 87.81$, whereas the PS of LSE-VCG has already reached 885.4. Overall, the maximum platform surplus achievable by LSE-VCG auction amounts to 989.03, compared to 674.85 for LS-VCG auction. Consequently, the LSE-VCG auction not only reaches budget surplus at an earlier stage but also consistently yields superior outcomes in terms of platform's surplus, social welfare, as well as rate of swap when compared to the LS-VCG auction under identical conditions.

To sum up, the escrow method significantly influences the LSE-VCG auction's realized social welfare, the platform's surplus, the payoffs to the agents, and the swap ratio of meta-items. By establishing an appropriate scale, the LSE-VCG auction ensures budget balance and enhances both social welfare and the frequency of meta-item trades in comparison to LS-VCG auction and any other standard auctions.

The outcomes presented have been derived from a context of 80 players and 3 atomic bids per XOR bid. Moving forward, this paper will assess how these outcomes vary with different quantities of atomic bids and agent numbers in the auction.

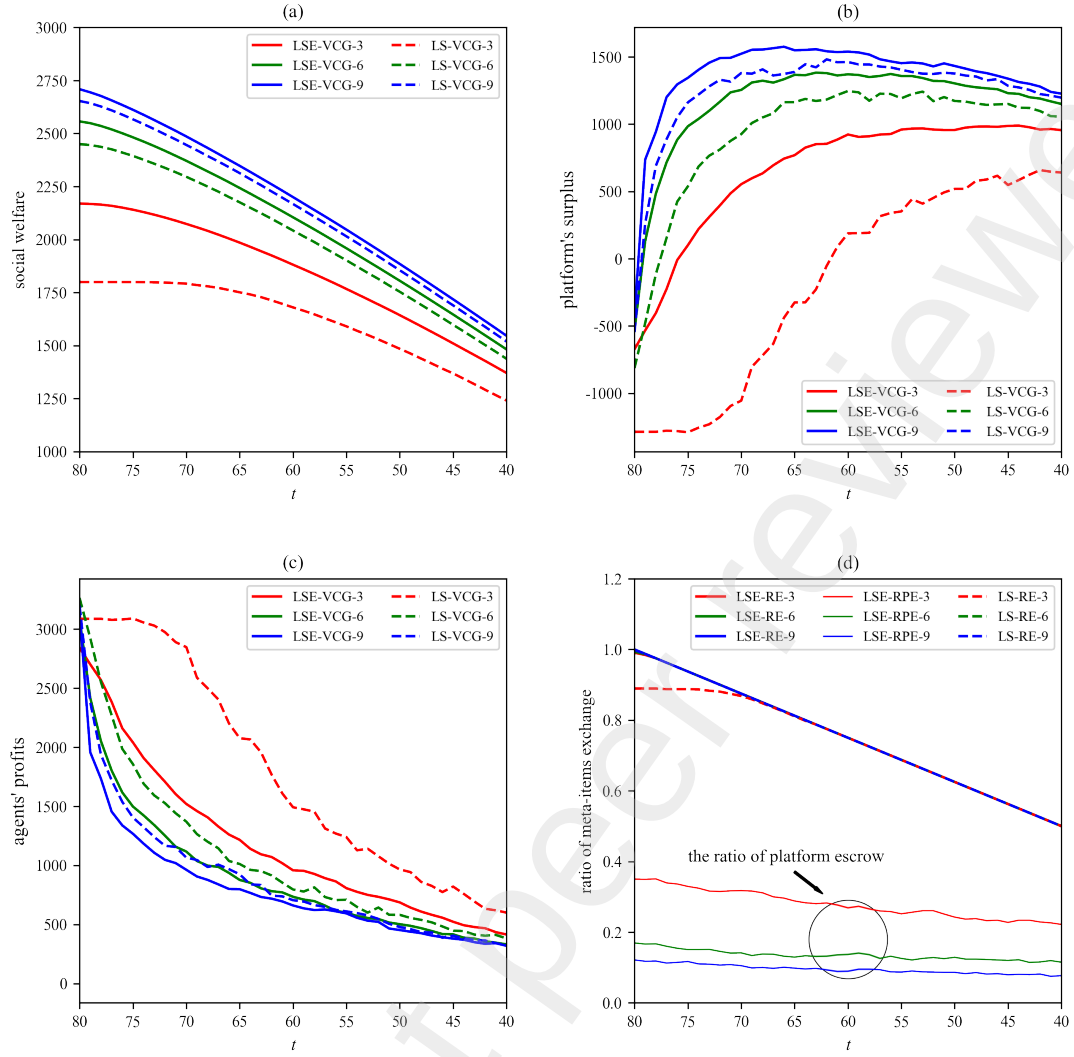


Fig. 7. The effect of the count of atomic bids

5.2.2 The effect of the count of indivisible bids

Fig. 7 examines how the number of indivisible bids in LSE-VCG and LS-VCG auctions affects five key performance metrics. In panels (a), (b), (c), we compare the two auction types: solid lines represent LSE-VCG auction, and dashed lines represent LS-VCG auction. The basic test with 3 indivisible bids per XOR bid is depicted by red curves, green curves indicate 6 indivisible bids, and blue curves represent 9 indivisible bids. Each curve is labeled with the method used for allocation and the count of indivisible bids; such as 'LSE-VCG-6' denotes LSE-VCG auction where each bid consists of 6 separate bids. Panel (d) shows the rates at which agents swap meta-items and meta-items they escrow in LSE-VCG auction.

Panel (a) depicts the social welfare across different values of t . A consistent trend is observed where the social welfare for both the LSE-VCG and LS-VCG auctions rises as the count of indivisible bids increases. Furthermore, the gap in social welfare between the LSE-VCG and LS-VCG auctions narrows with an increasing count of indivisible bids.

Panels (b) and (c) present the platform's surplus and agents' utility for both LSE-VCG and LS-VCG auctions. Notably, as the count of indivisible bids rises, agents' utility tends to

diminish, whereas the platform's surplus boosts. This inverse relationship may stem from a low substitutability among agents when fewer indivisible bids are present. Under a VCG-like pricing scheme, the market contributions of diverse agents overshadow those of substitutable ones. Additionally, it's observed that the limited supply threshold required for budget balance is increasing in the count of indivisible bids. This might be attributed to a smaller number of bids narrows down the potential traders pool, thereby requiring a reduced t for effectively managing the trade volume to maintain budget balance. Conversely, to achieve satisfactory profit levels for the agents, a higher t is suggested for markets with a higher indivisible bids number.

Conclusively, panel (d) illustrates the swap ratios of meta-items and the escrow ratio with the platform. In the LSE-VCG auction, the platform's escrow rates exhibit a declining trend with larger number of indivisible bids. Moreover, for values of $t < 78$, the LSE-VCG auction yields nearly identical swap rates across the all three bid scenarios; for values of $t < 69$, both the LSE-VCG and LS-VCG auctions show similar swap rates across all scenarios.

5.2.3 The effect of the count of agents

Fig. 8 examines how the count of participants influences four key performance indicators. The x-axes in this figure's panels display the difference between the total count of players and the limited supply parameter, that is $|N| - t$, as we aim to illustrate the impact of a consistent reduction in scale. For instance, we explore the consequences of reducing the scale by decrements of 10 agents. This scenario is represented by setting t to 30 for 40 agents, 70 for 80 agents, and 110 for 120 agents. Panels (a) through (d) compare the LSE-VCG and LS-VCG auctions, with solid lines indicating the LSE-VCG auction and dashed lines indicating the LS-VCG auction. The original simulation with 40 agents is represented by red curves, green curves for 80 agents, and blue curves for 120 agents. Each curve is labeled with the auction type and agent count; for example, 'LSE-VCG-80' denotes the LSE-VCG auction with 80 agents. Panel (d) shows the rates of total swaps in the both LSE-VCG and LS-VCG auctions, and ratio of platform escrow in the LSE-VCG auction.

Panel (a) indicates that both the LSE-VCG and LS-VCG auctions generate higher social welfare as the agents' number rises. Consistent with our predictions, increased market participation correlates with greater welfare generation.

Panel (b) shows that the platform's surplus of LSE-VCG auction rises with an increasing number of agents. The escrow approach ensures budget balance, and the threshold $|N| - t$ value increases with a larger number of agents. Moreover, even with 120 agents, budget surplus in LSE-VCG auction can be easily achieved by setting $|N| - t \geq 2$.

Panel (c) reveals that players' payoffs in both the two auctions rise with an increasing number of players. Additionally, the $|N| - t$ threshold at which the payoff curves of LSE-VCG and LS-VCG auctions intersect rises as the agent count grows. It is also noted that LS-VCG auction tends to yield higher agent payoffs than the LSE-VCG auction across all three scenarios.

Finally, in panel (d), the rates of item swaps under both two auction mechanisms largely overlap, while the platform escrow ratio for LSE-VCG gradually decreases from 0.2. Also note that the platform's valuations and agents' valuations of each other's meta-items follow the same

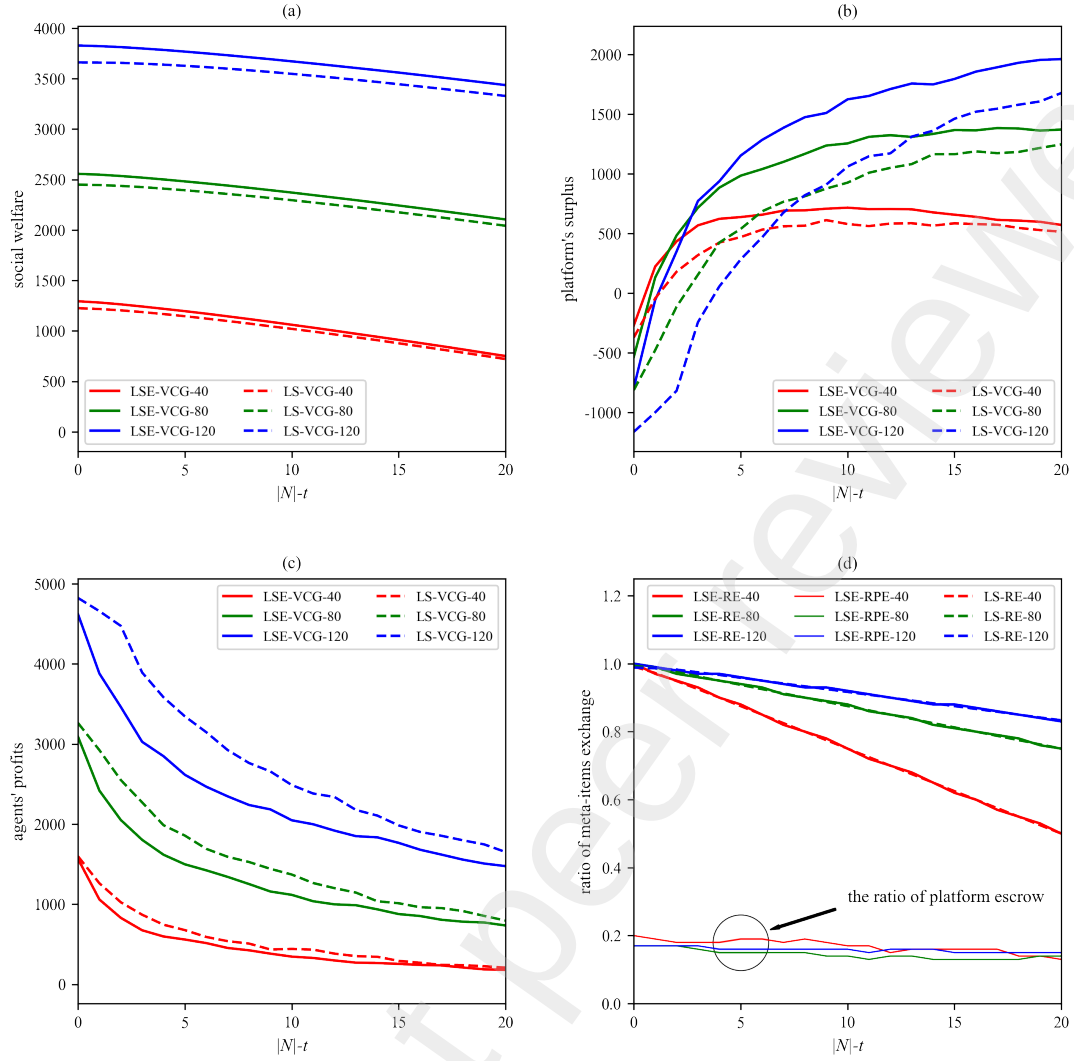


Fig. 8. The effect of the number of agents

distribution, resulting in a relatively stable selection rate for platform escrow.

5.2.4 The effect of the platform's bidding valuation fluctuations

Fig. 9 modifies the setup of the LSE-VCG auction. Previously, the platform's valuation was set to have the same price distribution function as the agents. Here, we have adjusted the platform's real valuations to 80% and 120% of the original.

In panel (a), social welfare increases with the rise of the platform's price, which is in accordance with common sense. In panel (b), the platform's surplus is also directly proportional to the platform's valuations. Panel (c) shows that agents' profits are inversely proportional to the platform's valuations, indicating that as the platform's valuations rise, the overall social welfare and platform's surplus increase, which is achieved at the expense of agent's profits. In panel (d), the overall ratio of meta-item swap is independent of the platform's valuations, while the platform's escrow ratio is directly proportional to the level of valuations.

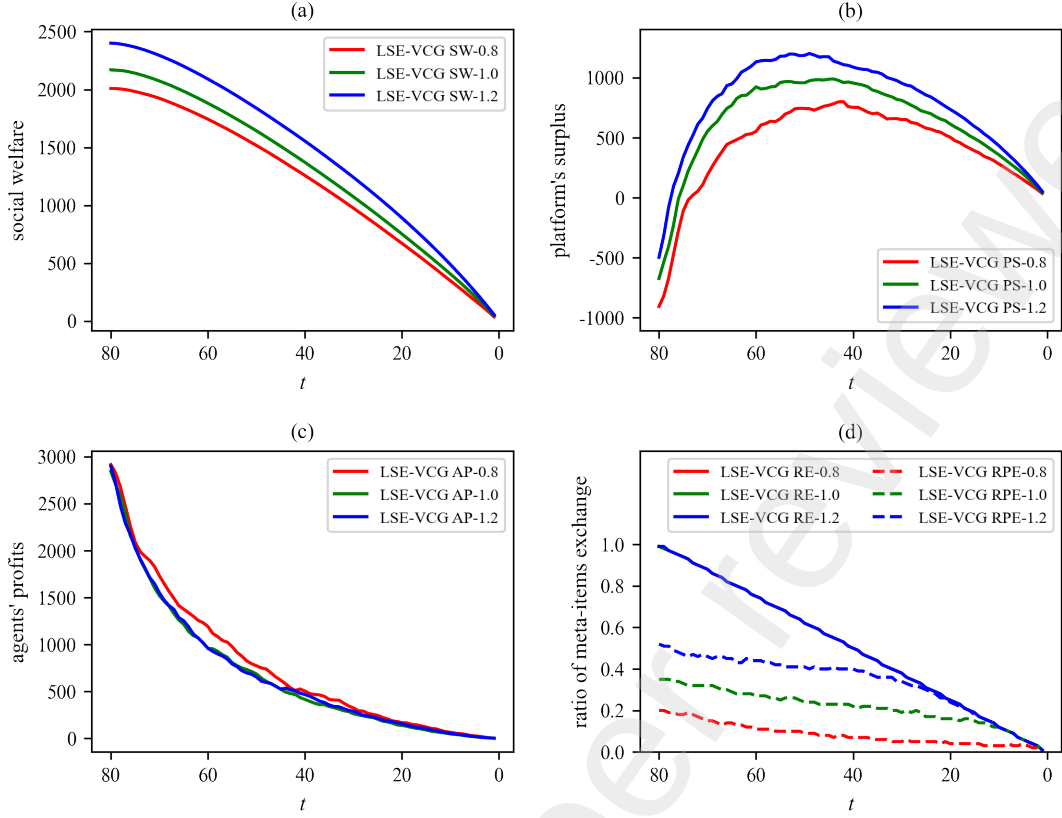


Fig. 9. The effect of the platform's bidding valuation fluctuations

5.2.5 The effect of platform's selfish telling

In Fig. 10, “selfish” refers to the platform's strategy of intentionally undervaluing or overvaluing its values for selfish gain. Specifically, while the platform's real valuations remain constant, he introduces false valuations at 80% and 120% of the actual price, which can significantly influence the overall auction process.

In panel (a), the social welfare is slightly higher when the platform engages in truthful telling compared to selfish telling. In panel (b), the platform's surplus is maximized through truthful telling. Theoretically, it is not always optimal for the platform to tell the truth. But in practice, due to the risks associated with over valuation or under valuation, truthful telling remains a good strategy. Moreover, the experiments also indicate that truthful telling is almost the optimal strategy for the platform. When the platform overstates the price rather than understates it, the platform's surplus is higher, likely due to the increased escrow rate caused by the platform's overvaluation. In panel (c), undervaluing through selfish telling leads to more meta-item trades among agents, resulting in the highest agents' profits. Conversely, agents' benefits are lower when the platform overvalues through selfish telling. In panel (d), the overall transaction ratio of the auction is independent of the platform's telling valuation behavior. The platform's escrow rate is directly proportional to its telling valuation, which confirms the results from the previous experiment.

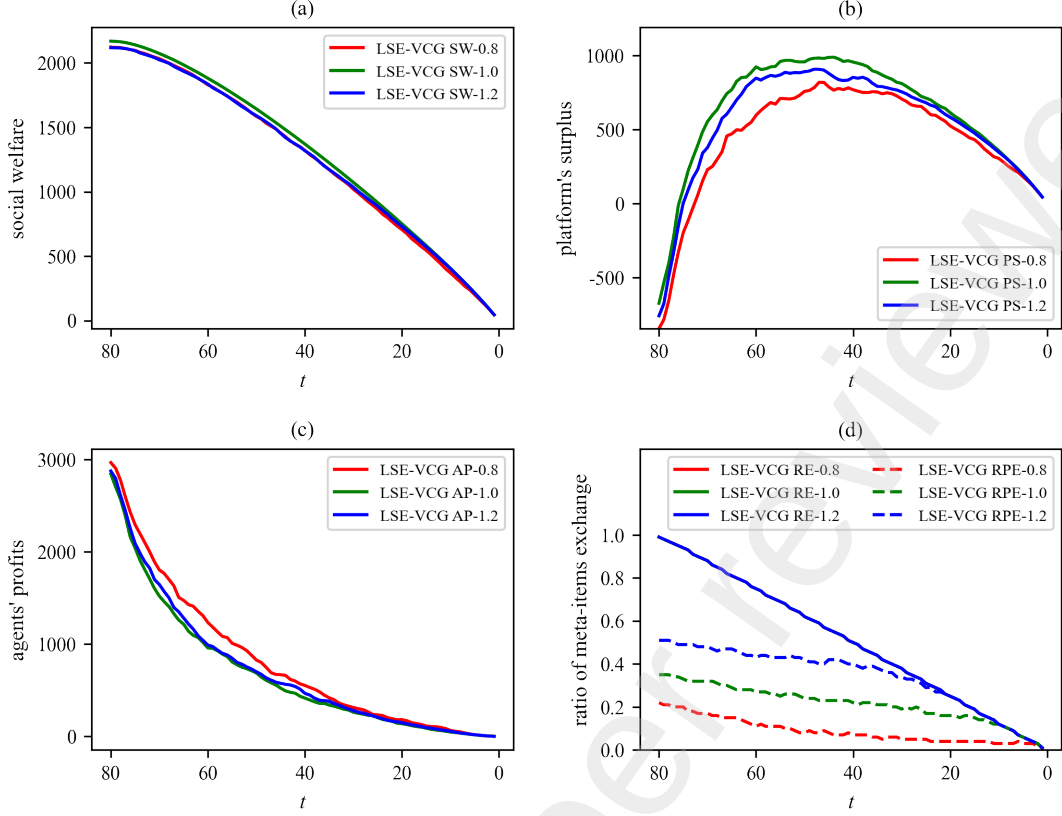


Fig. 10. The effect of platform's selfish telling

5.2.6 The effect of auction batch size

Fig. 11 presents the experimental results for the LSE-VCG auction tests under various batched auction configurations. The scenarios include a one-shot auction with all 120 agents in a single batch, as well as batched auctions divided into 3, 6, and 12 batches.

In panel (a), the one-shot auction achieves the highest total social welfare, and social welfare declines as the number of auction batches increases. Panel (b) shows a similar trend, with the platform's surplus peaking in the one-shot auction and diminishing in batched auctions, particularly with the highest number of batches. However, it is interesting that batched auctions can achieve budget balance earlier than the one-shot auction as the limited supply number decreases from 120 to 113.

In panel (c), when $t > 100$, the one-shot auction yields the highest agents' profits, while batched auctions result in lower profits, decreasing as the count of batches increases. This trend is consistent with panel (b). When $t < 100$, the one-shot auction results in the lowest agents' profits, with slightly higher profits in batched auctions and similar impacts of batch sizes. This suggests that the benefits of one-shot and batched auctions are primarily reflected in the platform's surplus when $t < 100$. Panel (d) indicates that the platform's escrow rate increases with batch size in the batched auctions. The overall transaction ratio is nearly identical across all configurations.

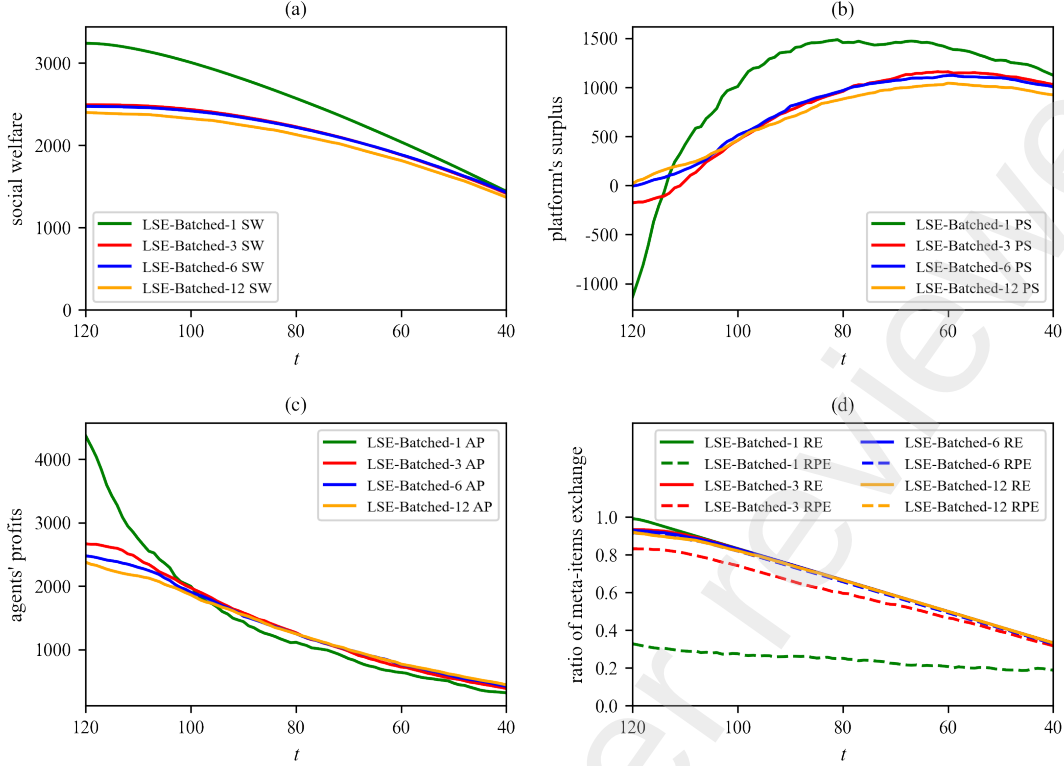


Fig. 11. The effect of auction batch size

6 Conclusions

In this paper, we propose effective VCG-like auction mechanisms for metaverse trading environments. We consider a metaverse multilateral swap setting with a third-party platform/auctioneer (e.g., Foundation NFT company) and multiple agents (bidders). Each agent is simultaneously a buyer and a seller of virtual assets. To prevent the platform's budget deficit, we implement limited supply (Chu and Shen, 2007) which limits the upper bound on the quantity of winners, and introduce a platform escrow method. Through the platform escrow method, each agent can sell the usage right of his meta-item to the platform and receive a corresponding VCG payment.

This study extends the existing auction theories. By introducing both limited supply and platform escrow, we propose a novel VCG-like auction mechanism. This auction mechanism has good properties, including incentive compatibility and individual rationality. The proposed efficient auction mechanisms ensure budget balance and attempt to maximize the total social welfare. We prove that if the multilateral matching achieves maximal social welfare and the number of swaps is $|N|$, where $|N|$ is the number of bidders, then the *buyer substitute* condition does not hold (called *impossibility theorem*, i.e., Theorem 1). Due to the favorable properties of limited supply, we further prove that the platform's revenue is weakly larger if the maximum number of winners is equal to $|N| - 1$.

Furthermore, we conduct some experimental simulations for the VCG auction with limited supply (LS-VCG auction) and our mechanism with limited supply and platform escrow (LSE-VCG auction). We also employed a machine learning method to forecast the limited supply number. We made predictions under two objectives: achieving maximum platform surplus and

minimum positive platform surplus. Some intriguing results and insights are as follows:

- The LS-VCG auction and LSE-VCG auction could reach greater social welfare, agents' profits and ratio of swap than standard auctions, such as English auction, Dutch auction, the first-price sealed auction and Vickrey auction.
- In contrast to the LS-VCG auction, the LSE-VCG auction we propose shows better performance with regard to social welfare and the ratio of successful tradings, as well as the platform's surplus.
- While theoretical analysis suggests that the platform could gain from a selfish telling, our simulation results indicate that truthful telling is nearly optimal. The result contrasts with the traditional combinatorial VCG auctions. The discrepancy can be attributed to the risks of overstating or understating in the context of multilateral swaps.
- Conducting batch auctions can achieve budget balance earlier compared to a one-shot auction as the limited supply number decreases. Furthermore, to maintain budget balance, a higher limited supply number is required with an increased frequency of auctions.
- We demonstrate the effectiveness of platform escrow in extensive experimental environments. Thus, the platform should properly combine the schemes of limited supply and platform escrow to enhance budget surplus.
- The prediction for achieving maximum platform surplus can lead to high platform revenue, and the prediction for minimum positive platform surplus can attract more participants.

The key managerial implications of our findings are as follows: First, the design of metaverse swap mechanisms can be based on the underlying idea of VCG auctions, as well as the schemes and machine learning methods that can achieve budget balance. Second, the platform needs to tell the truth for enhancing revenue and attracting more agents in the metaverse market. Third, conducting auctions more frequently enables earlier achievement of budget balance compared to infrequent auctions, as the limited supply parameter diminishes. Increased auction frequency necessitates a higher limited supply value to attain budget surplus. Finally, when designing metaverse swap mechanisms, the platform (auctioneer) should have a full understanding of the benefits and risks related to metaverse float pricing problems.

Looking ahead to future works, our research can be expanded in several directions. Firstly, one might try more sophisticated machine learning methods such as neural pseudogradient ascent (Bichler et al., 2023) and feedback-based dynamic mechanism (Daskalakis and Syrgkanis, 2022) to forecast the optimal limited supply number in more general settings. Secondly, since the common value of a metaverse item can increase with the number of traders in the market, the impact of (positive) network effects can also be incorporated in our auction model. Thirdly, integrating a reserve price mechanism into the VCG auction framework may enhance the platform's earnings (Bhargava et al., 2020), which would help address budgetary concerns – this can be contrasted with our current method. Establishing metaverse assets auction mechanisms has a huge potential in utilizing the digital resource and expanding the metaverse marketplace. We are convinced that further exploitation and exploration in this domain are necessary.

Notes

The interview summary and audio recordings are available upon request. The statistical results of the questionnaires are put in supplemental materials.

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